

application of derivatives

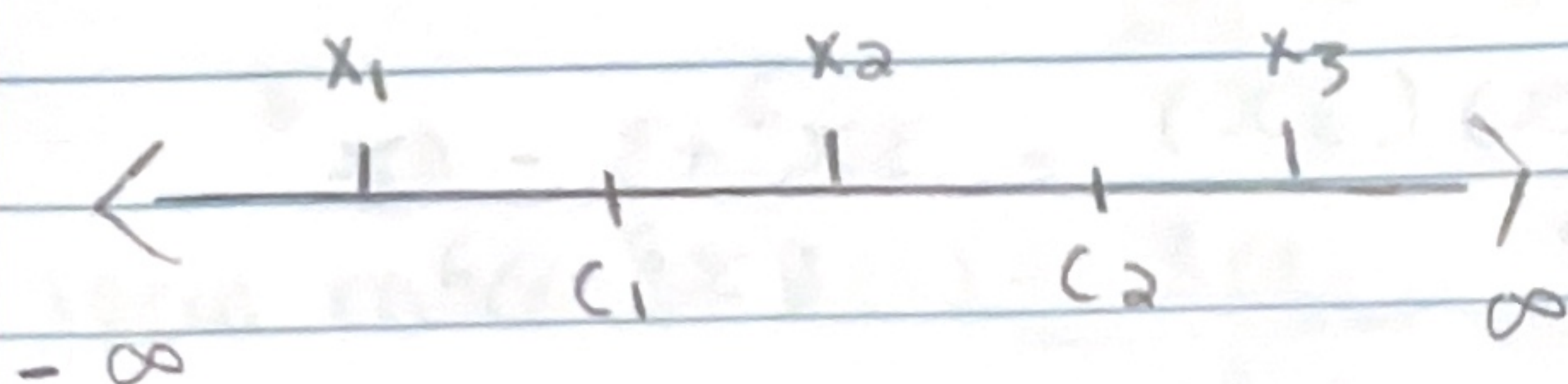
Step ①: Find the domain of $f(x)$

Step ②: Find the derivative of $f(x)$

Step ③: $f'(x) = 0$ to find critical number

Step ④: $f'(x) = DNE$ (for this step you need to set the denominator to zero because when it is set to zero it does not exist)

(you must do step 3 and 4)



c_1 = first critical number

c_2 = second critical number

these are always going to be two also place them in order and if one of the critical points is not in the denominator it cannot be a critical number

x_1, x_2, x_3 : these are points you pick in between the interval and test them to see if you get a positive or negative (increase or decrease)

$(-\infty, c_1) = f'(x_1) =$ if $(+)$ increasing or if $(-)$ decreasing

$(c_1, c_2) = f'(x_2) =$ if $(+)$ increasing or if $(-)$ decreasing

$(c_2, \infty) = f'(x_3) =$ if $(+)$ increasing or if $(-)$ decreasing

Example 1:

$$f(x) = \frac{3x}{x^2+1} + \frac{6}{5}$$

Domain: $(-\infty, \infty)$

$$f'(x) = \frac{-3x^2+3}{(x^2+1)^2}$$

$$f = 3x \quad g = x^2+1 \quad \frac{df}{dx} \cdot g - f \cdot \frac{dg}{dx}$$

$$\frac{df}{dx} = 3$$

$$\frac{dg}{dx} = 2x$$

$$g^2$$

$$\frac{(3)(x^2+1) - (3x)(2x)}{(x^2+1)^2} = \frac{3x^2+3-6x^2}{(x^2+1)^2}$$

$$\Rightarrow \frac{-3x^2+3}{(x^2+1)^2}$$

$$f'(x) = 0$$

$$\frac{-3x^2+3}{(x^2+1)^2} = 0$$

$$-3(x^2-1) = 0$$

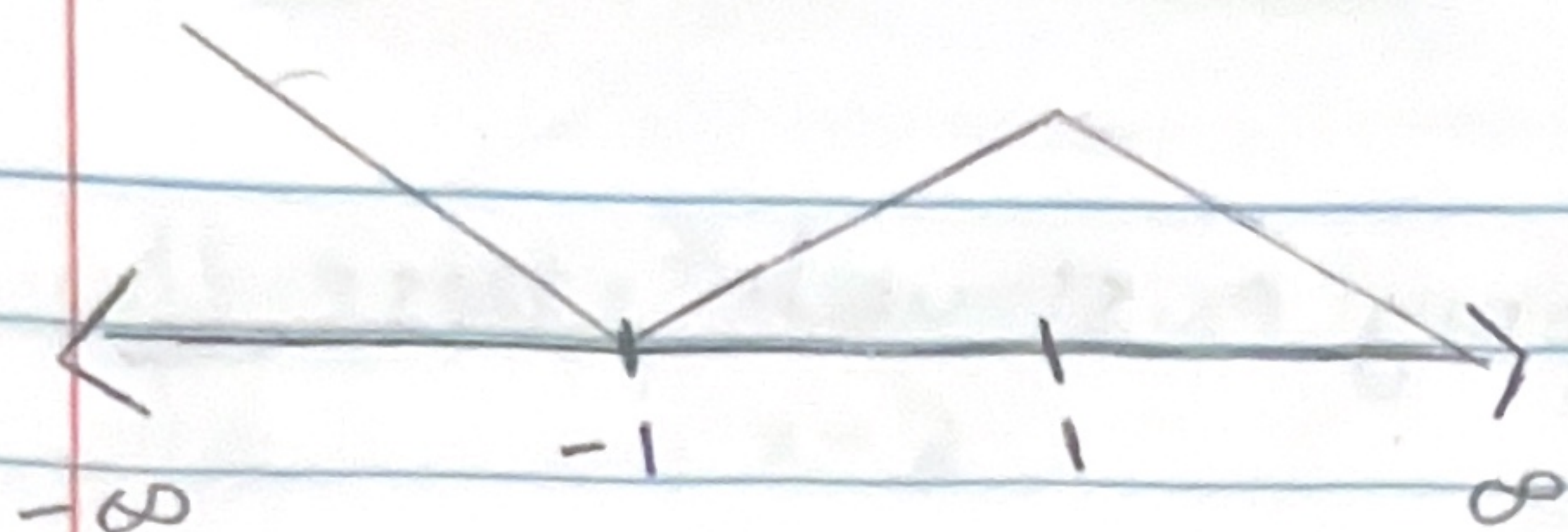
$$x = 1, x = -1$$

$$f'(x) = DNE$$

$$\sqrt{(x^2+1)^2} = \sqrt{0}$$

$$x^2+1 = 0$$

$$x^2 \neq -1$$



$$(-\infty, -1) = f'(-3) = \frac{-3(-1)^2 + 3}{((-1)^2 + 1)^2} = -0.24$$

$$(-1, 1) = f'(0) = \frac{-3(0)^2 + 3}{((0)^2 + 1)^2} = 3$$

$$(1, \infty) = f'(2) = \frac{-3(2)^2 + 3}{((2)^2 + 1)^2} = -0.36$$

$$\text{local min} = f(-1) = \frac{3(-1)}{(-1)^2 + 1} + \frac{6}{5} = -\frac{3}{10}$$

$$\text{local max} = f(1) = \frac{3(1)}{(1)^2 + 1} + \frac{6}{5} = \frac{27}{10}$$

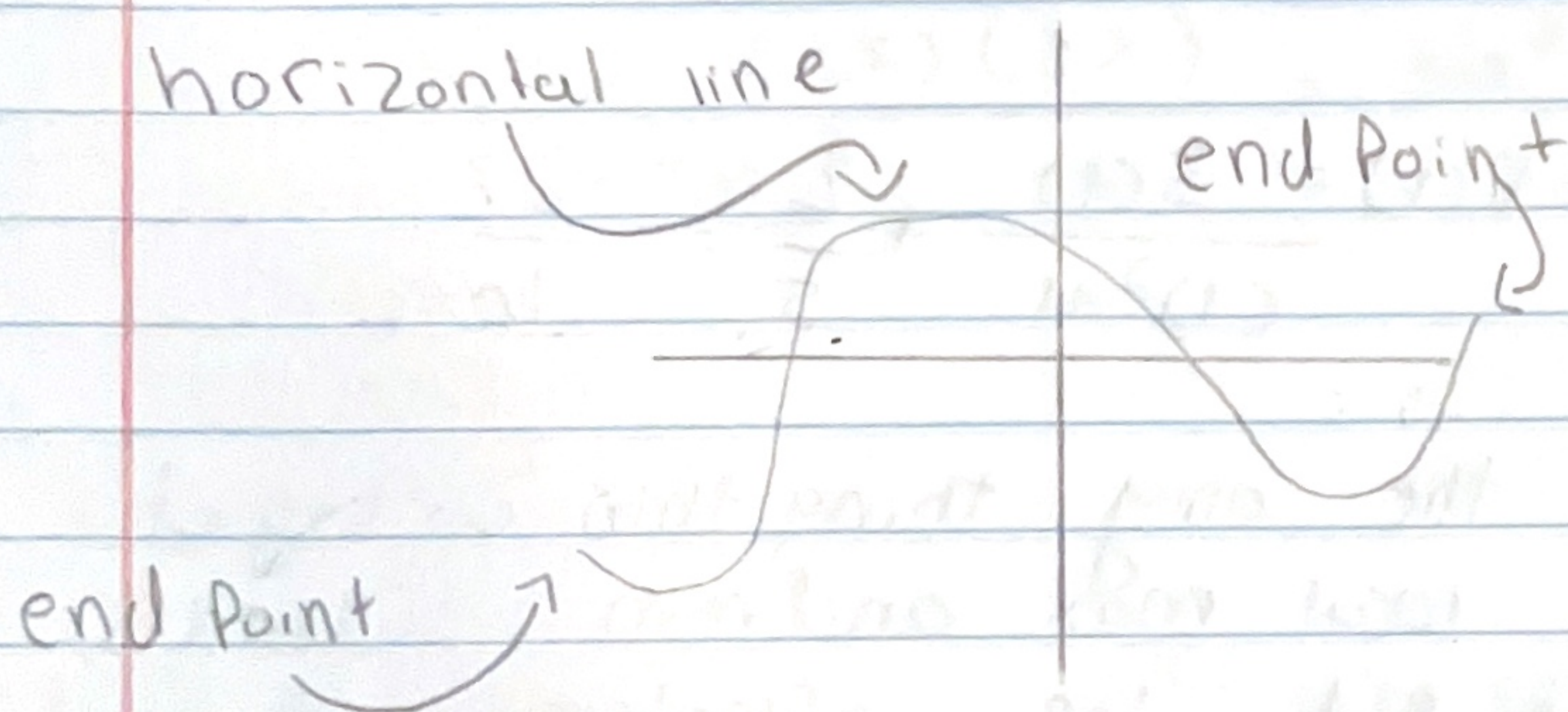
up to this the only thing this will get you is the local max and min if there are any to get the inflection point and concave up and down you must find the second derivative.

a critical number only exist under these conditions

$f(a) = \text{exist}$
 $f'(a) = 0$ } then a is a critical number

$f(a) = \text{exist}$
 $f'(a) = \text{DNE}$ } then a is a critical number

end point do not qualify as critical points because we do not know what happens after that



horizontal lines have a slope of zero

the first derivative was given

$$\frac{df}{dx} = \frac{2x^2 + x - 10}{x - 2}$$

$$\text{Domain} = (-\infty, 2) \cup (2, \infty)$$

$$x - 2 \neq 0$$

$$x \neq 2$$

$$\frac{d^2f}{dx^2} = \frac{2x - 4}{(x - 2)}$$

you can simplify I just didn't to make it easier to find the critical numbers

$$f = 2x^2 + x - 10$$

$$g = x - 2$$

$$\frac{\frac{df}{dx} \cdot g - f \cdot \frac{dg}{dx}}{g^2}$$

$$\frac{df}{dx} = 4x + 1$$

$$\frac{dg}{dx} = 1$$

$$\frac{(4x + 1)(x - 2) - (2x^2 + x - 10)(1)}{(x - 2)^2}$$

$$f'(x) = 0$$

$$f'(x) = \text{DNE}$$

$$\frac{2x - 4}{(x - 2)} = 0$$

$$(x - 2) = 0$$

$$x = 2$$

$$2(x - 2) = 0$$

$$x = 2$$

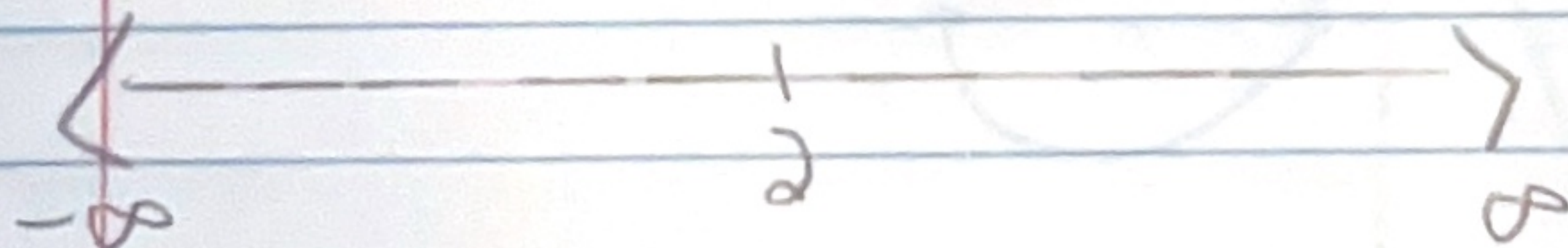
Not in domain

Not in domain

$$\frac{(4x + 1)(x - 2) - (2x + 5)(x - 2)}{(x - 2)^2}$$

$$\cancel{(x - 2)} \left[\frac{(4x + 1) - (2x + 5)}{(x - 2)^2} \right]$$

$$\frac{2x - 4}{(x - 2)} \quad \Leftarrow$$

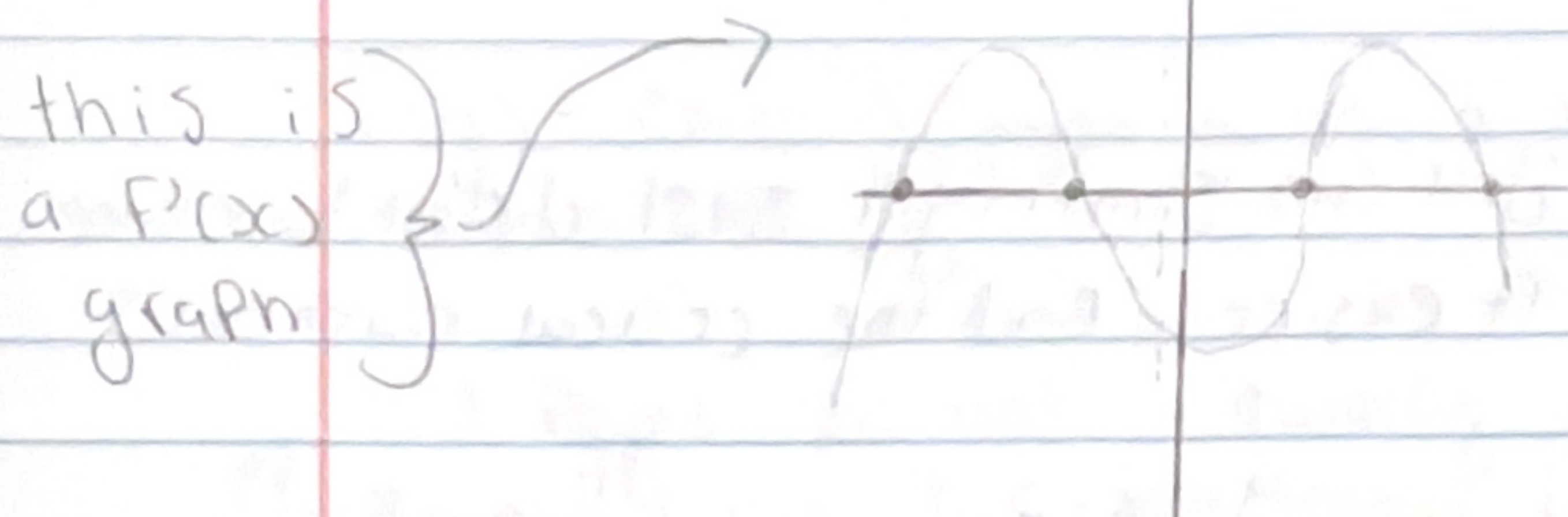


$$(-\infty, 2) = f''(1) \quad \frac{2(1) - 4}{1 - 2} = 2$$

$$(2, \infty) = f''(3) = \frac{2(3) - 4}{3 - 2} = 2$$

No concave up, down
No inflection points

- $f(x)$ critical numbers in $f'(x)$ graph are the x -intercepts

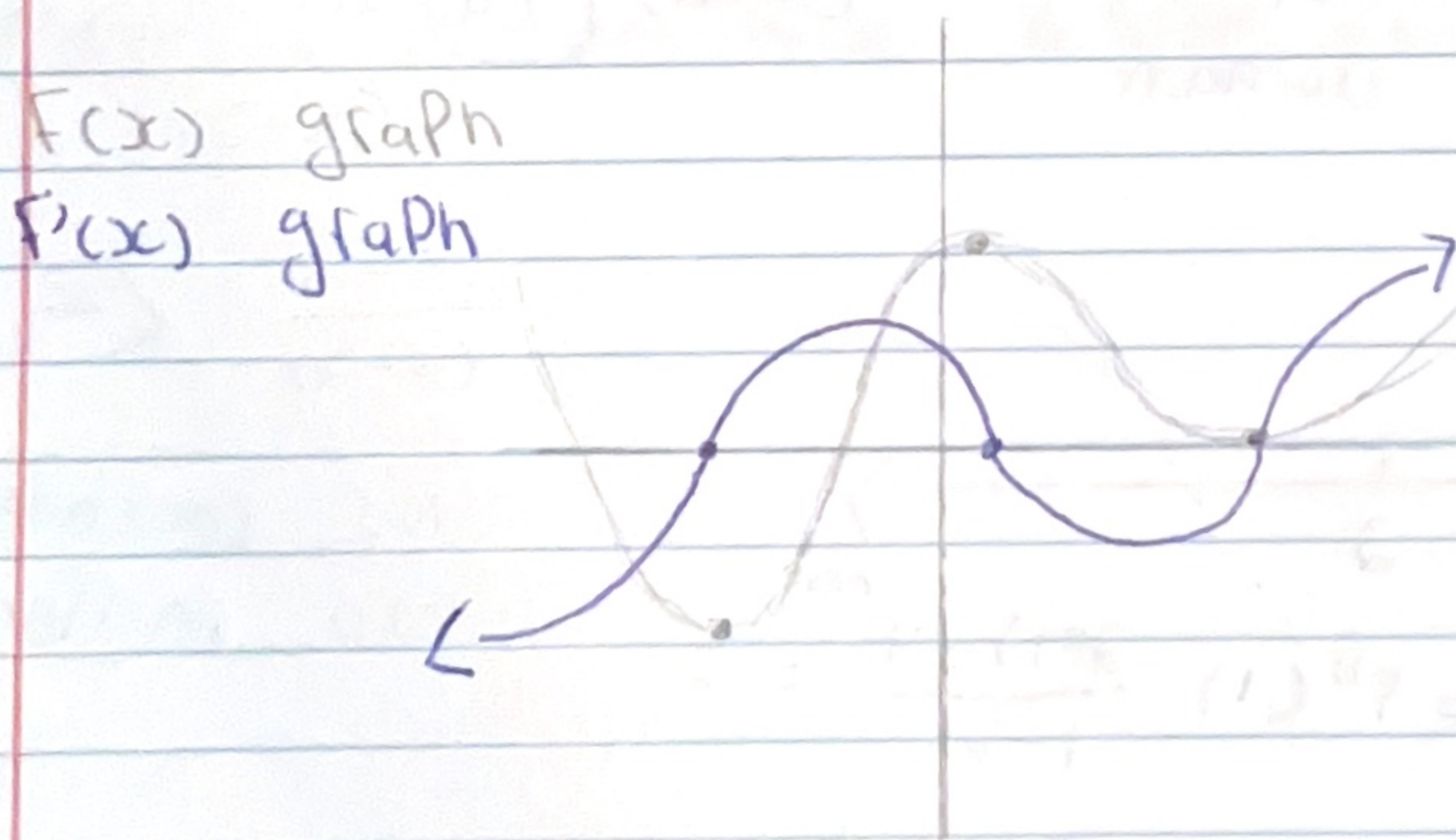


if they give you the $f'(x)$ graph and they ask for $f(x)$ critical number look at the x -int

- $f(x)$ is increasing ($f'(x) > 0$) $\rightarrow$ in the $f'(x)$ graph where it is above the x -axis

Visual:

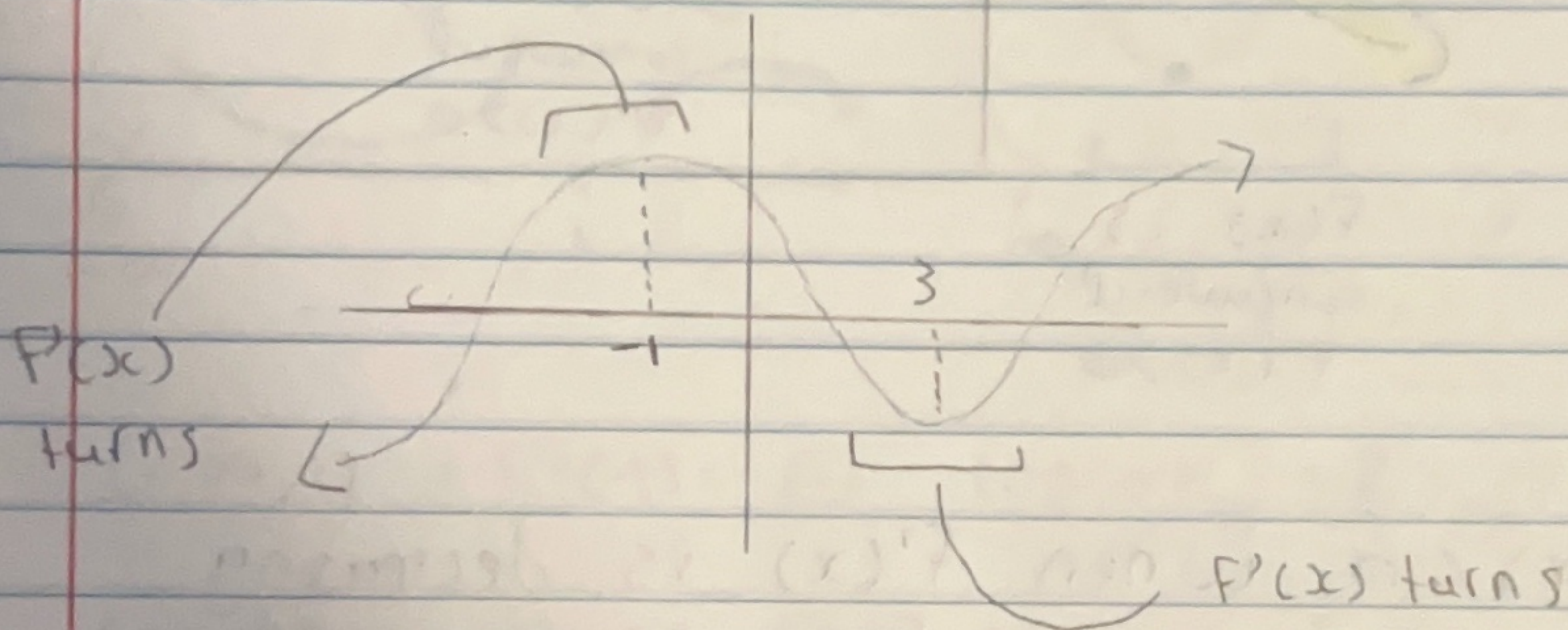
$f(x)$ is increasing where $f'(x)$ above x -axis



also if you look at the $f'(x)$ graph x -value in will be the critical point of $f(x)$

- $F(x)$ is decreasing where the $F'(x)$ ($P(x) < 0$) is below the x -axis

- $F''(x) = 0$ when $F'(x)$ turns

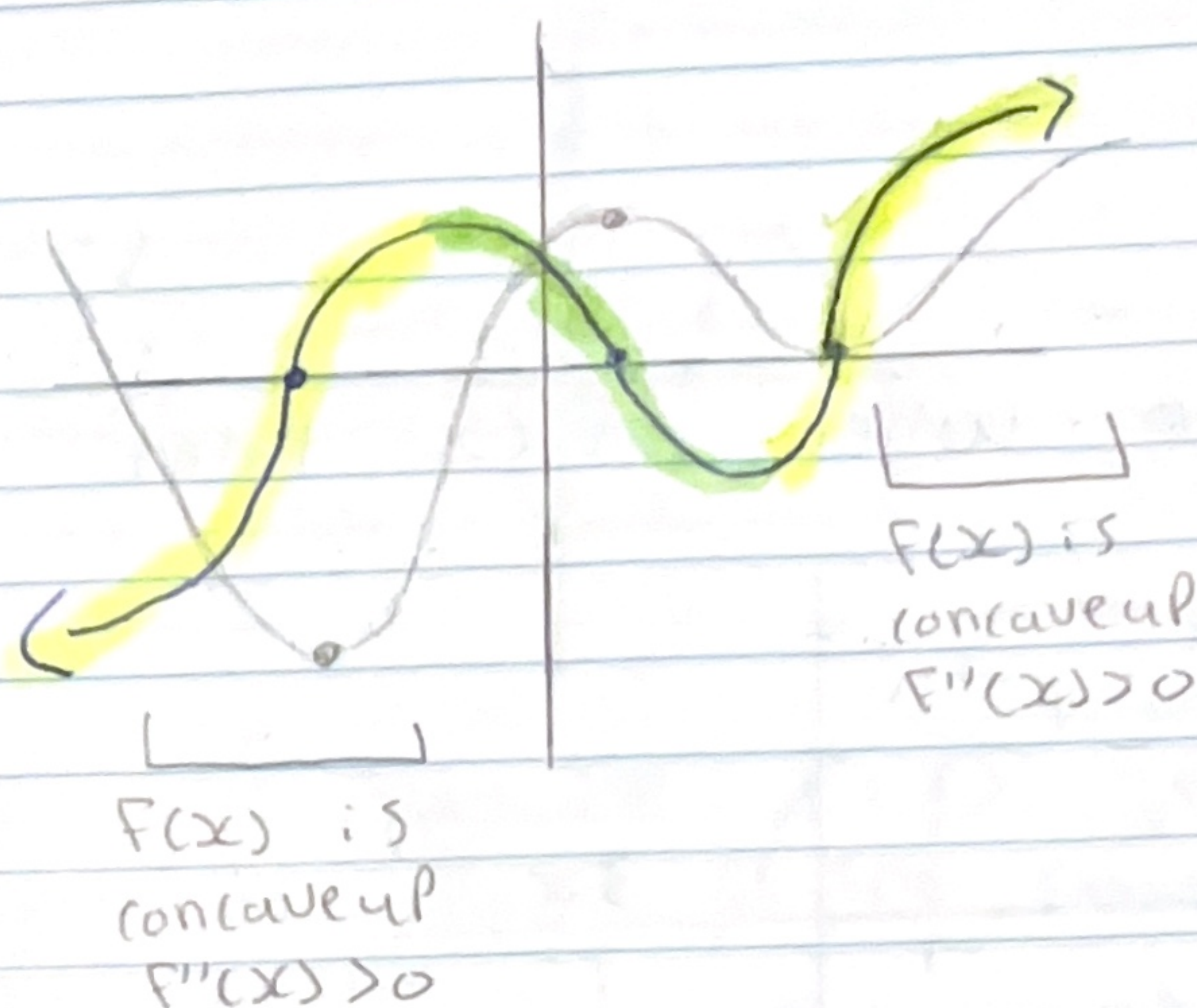


So $F''(x) = 0$ @ -1 and 3

$f''(x) > 0$ when $f'(x)$ is increasing
this also where $f(x)$ is concave up

$f'(x)$ in blue
 $f(x)$ in regular

$f''(x) > 0$



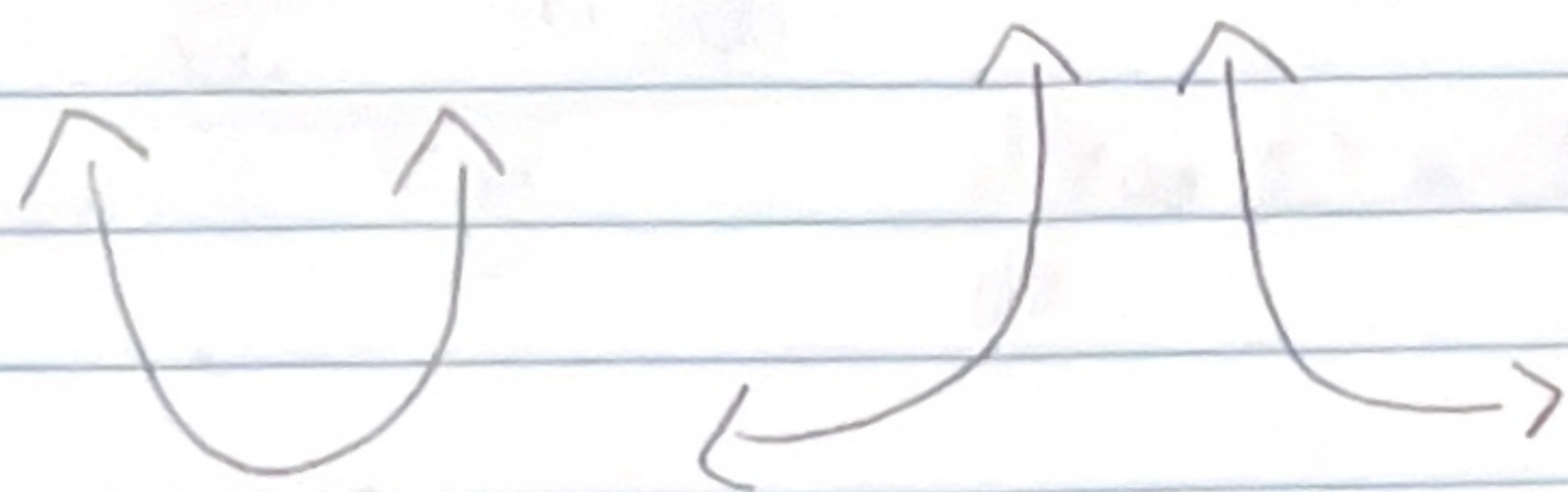
$f''(x) < 0$ when $f'(x)$ is decreasing
this also where $f(x)$ is concave down

$f''(x) < 0$

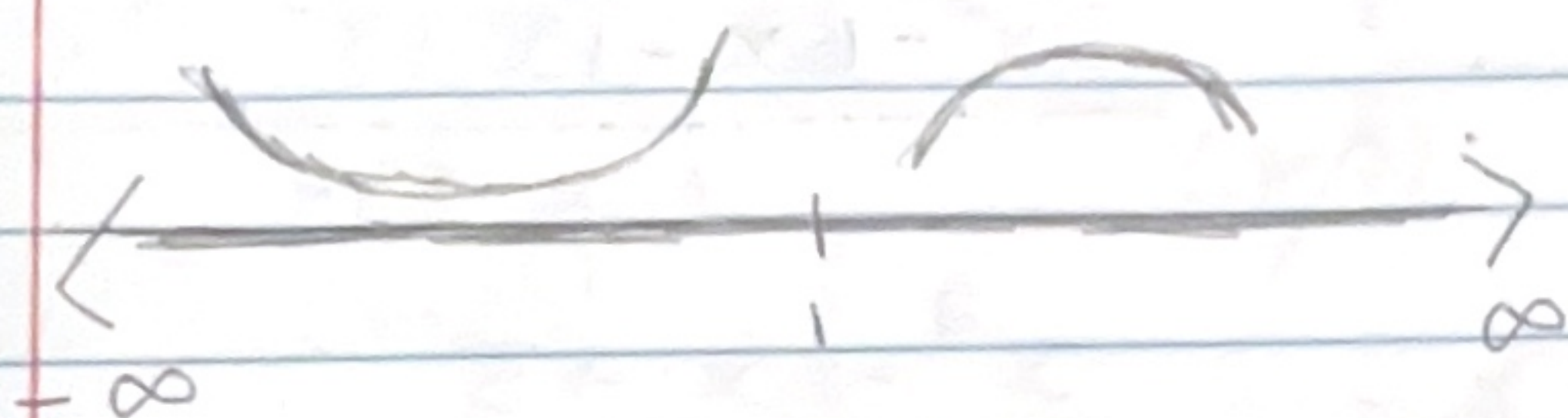
Concavity is not the local maximum or minimum. Concavity describes the curve while the local maximum and minimum tell us the highest on a small portion of the graph. That's why it's called local max or min.

Critical Points and inflection must be within the domain

if $f''(x) > 0$ then its concave up



this is what they would look like on the graph
you could not find the concavity by just looking
at the graph you must find the second derivative

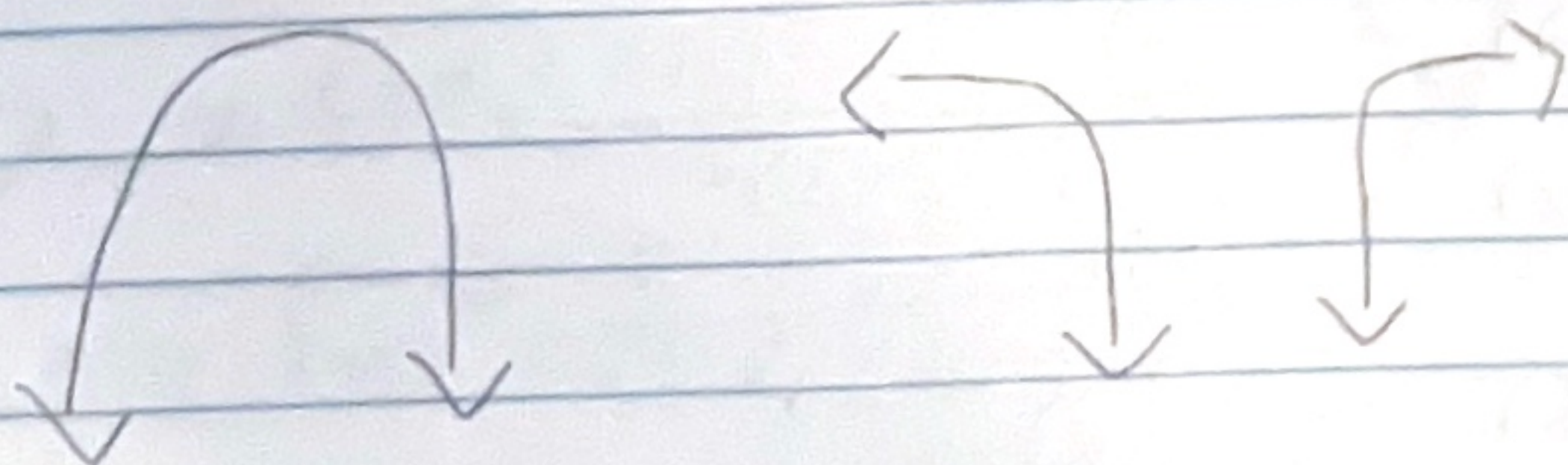


$(-\infty, 1) = f''(-1) = \ominus$ concave up

$(1, \infty) = f''(3) = \oplus$ concave down

the inflection point is where the change happened where it went up then down or down then up it doesn't really matter only thing that does is at which point for this example the inflection point is $x=1$

if $f''(x) < 0$ then its concave down



$$f(x) = \frac{x^2 - x - 2}{x - 3}$$

$$\frac{\frac{df}{dx} \cdot g - f \cdot \frac{dg}{dx}}{g^2}$$

Domain: $(-\infty, 3) \cup (3, \infty)$

$$x - 3 \neq 0$$

$$x \neq 3$$

$$f = x^2 - x - 2 \quad g = x - 3 \quad \frac{(2x-1)(x-3) - (x^2 - x - 2)}{(x-3)^2}$$

$$\frac{df}{dx} = 2x - 1 \quad \frac{dg}{dx} = 1$$

$$\frac{2x^2 - 6x - 1x + 3 - x^2 + x + 2}{(x-3)^2}$$

$$\frac{df}{dx} = \frac{x^2 - 6x + 5}{(x-3)^2}$$

$$\frac{x^2 - 6x + 5}{(x-3)^2}$$

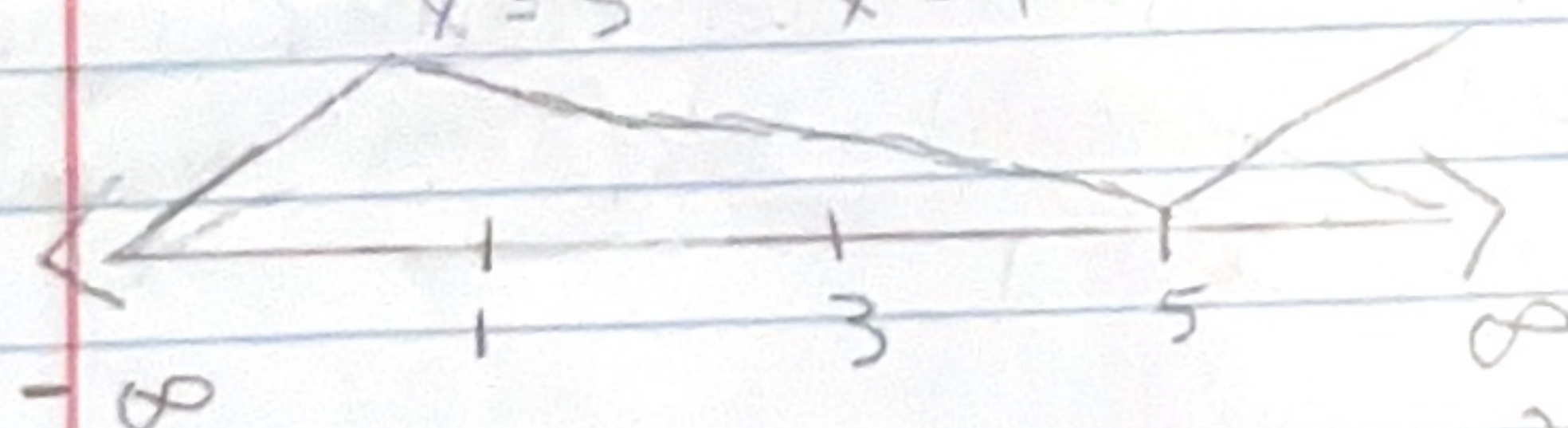
$$f'(x) = 0$$

$$\frac{x^2 - 6x + 5}{(x-3)^2} = 0$$

$$x^2 - 6x + 5 = 0$$

$$(x-5)(x-1) = 0$$

$$x = 5 \quad x = 1$$



$$f'(x) = \text{DNE}$$

$$\sqrt{(x-3)^2} = \sqrt{0}$$

$$x - 3 = 0$$

$$x \neq 3$$

cannot work because 3 is not in domain

$$f(1) = 1 \text{ local max}$$

$$f(5) = 9 \text{ local min}$$

$$(-\infty, 1) = f'(0) = \frac{(0)^2 - 6(0) + 5}{(0-3)^2} = \frac{5}{9}$$

$$(1, 3) = f'(2) = \frac{(2)^2 - 6(2) + 5}{(2-3)^2} = -3$$

$$(3, 5) = f'(4) = \frac{(4)^2 - 6(4) + 5}{(4-3)^2} = -3$$

$$(5, \infty) = f'(6) = \frac{(6)^2 - 6(6) + 5}{(6-3)^2} = \frac{5}{9}$$

$$\frac{df}{dx} = \frac{x^2 - 6x + 5}{(x-3)^2}$$

$$F = x^2 - 6x + 5$$

$$g = (x-3)^2$$

$$\frac{df}{dx} = 2x - 6$$

$$\frac{dg}{dx} = 2(x-3)$$

$$\frac{\frac{df}{dx} \cdot g - f \cdot \frac{dg}{dx}}{g^2} \Rightarrow$$

$$\frac{(2x-6)(x-3)^2 - (x^2-6x+5)(2(x-3))}{(x-3)^4}$$

$$(x-3) \left[(2x-6)(x-3) - 2(x^2-6x+5) \right]$$

$$(x-3) \left[(x-3)^3 \right]$$

$$\frac{d^2f}{dx^2} = \frac{8}{(x-3)^3}$$

$$2x^2 - 6x - 6x + 18 - 2x^2 + 12x - 10$$

$$\frac{8}{(x-3)^3}$$

$$F''(x) = 0$$

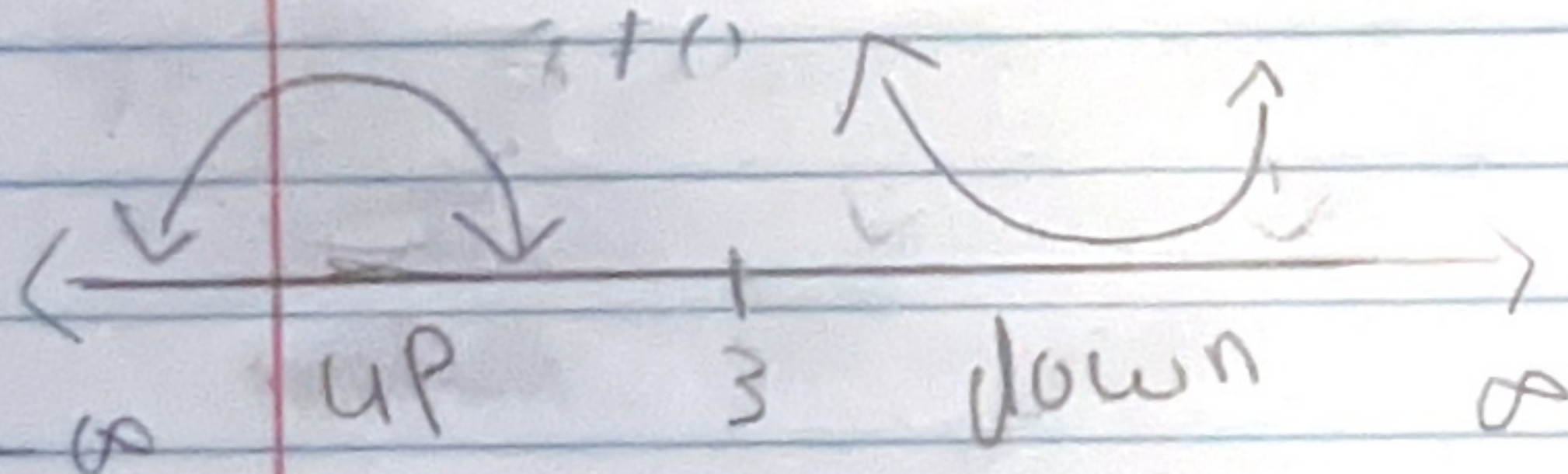
$$\frac{8}{(x-3)^3} = 0$$

$$F''(x) = \text{DNE}$$

$$\sqrt[3]{(x-3)^3} = \sqrt[3]{0}$$

$$x-3 = 0$$

$$x \neq 3$$



$$(-\infty, 3) = F''(0) = \frac{8}{(0-3)^3} = -\frac{8}{27} \text{ (concave down)}$$

$$(3, \infty) = F''(4) = \frac{8}{(4-3)^3} = 8 \text{ (concave up)}$$

$$\text{inflection Point} = x = 3$$

Even though you have not found critical points you still find a number to plug the three come from the domain